

# Velocity sorting biases the equilibrium in Array-RQMC for binary-collision models

James M. Hyman\*

August 19, 2026

## Abstract

Array-RQMC couples the states of many simulated trajectories by sorting them at each step and aligning the sorted states with a low-discrepancy point set, and it has produced large variance reductions across a range of Markov chain problems. We consider one way of using it for binary-collision models of the Kac type: ordering the particles of a system by their own state and letting that order choose which particles collide. We show that this adaptation cannot work. For a Kac model on the sphere  $\sum_i v_i^2 = N$  with  $N$  even, averaging one collision over the scattering angle gives a fourth-moment identity in which the pairing rule enters only through the single scalar  $C(P, v) = \sum_{(a,b) \in P} v_a^2 v_b^2$  determined by the matching  $P$ . The equilibrium value of this scalar is attained exactly by pairings chosen independently of the state, and by a rearrangement argument it is maximised by pairing partners of equal speed and minimised by pairing extremes. Hence any pairing whose cross term departs from its exchangeable value moves the fourth moment on the first collision, and the equilibrium distribution is not preserved, for every even  $N \geq 4$ . The identity extends to any symmetric scattering law common to all pairs, with the drift attenuated by a computable factor, so small-angle scattering does not avoid the conclusion. Within this adaptation the alignment that produces the variance reduction is the same structure that biases the equilibrium, so no sorting key separates them. The obstruction is to the adaptation, not to Array-RQMC itself: a construction that sorts independent copies of the chain and leaves each copy's collision partners alone is untouched by it. In numerical experiments the variance reduction is delivered as advertised while the mean squared error is monotonically worsened, the second moment remains exact to machine precision while the fourth is wrong by a factor of several hundred, and replicates agree closely on an incorrect answer. The proof supplies a one-pass  $O(N)$  diagnostic that evaluates a proposed pairing before any trajectory is run, and identifies which sorting strategies remain admissible.

**Key words.** randomized quasi-Monte Carlo, Array-RQMC, Kac model, invariant measure, particle methods, collisional kinetics

**AMS subject classifications.** 65C05, 65C35, 82C40, 11K45

---

\*Department of Mathematics, Tulane University, New Orleans, LA 70118 ([mhyman@tulane.edu](mailto:mhyman@tulane.edu)).

# 1 Introduction

Array-RQMC, introduced by L’Ecuyer, Lécot and Tuffin [9] and anticipated for Markov chains in one dimension by Lécot and Tuffin [8], simulates many trajectories together rather than one at a time. At each step the states are sorted, the sorted states are aligned with a low-discrepancy point set, and every trajectory is advanced using its assigned point. The method has produced substantial variance reductions across a range of problems, and the empirical study of sorting rules by L’Ecuyer et al. [10] showed that the choice of sorting key matters a good deal, with the Hilbert-curve ordering of Gerber and Chopin [3] performing well in several dimensions. Related work on space-filling curves for sampling appears in He and Owen [4], and recent applications to grid-free solvers in Ho and Owen [5, 6], where an Array-RQMC walk on spheres with a Hilbert sort reduces the variance by factors running from tens to thousands. General background on randomized quasi-Monte Carlo is in Niederreiter [12], Dick and Pillichshammer [2] and Owen [13].

Part of why the method works is understood. Randomized quasi-Monte Carlo is most effective when the integrand is dominated by the low-order terms of its ANOVA decomposition, a property made precise through the effective dimension of Caflisch, Morokoff and Owen [1] and the mean dimension of Liu and Owen [11]; sorting the states is what allows a chain to present the point set with a low-dimensional integrand at each step.

Collisional kinetics looks like a natural target for the method. The trajectories are long, the per-step dimension is small, and the quantity of interest is an average over many particles: the regime in which Array-RQMC has done well elsewhere. The obvious sorting key involves the particle velocities, since it is the velocities that the collision operator acts on.

We show that this adaptation cannot work, and that the reason is structural rather than a matter of implementation. Section 4 establishes an identity for a Kac binary-collision model: averaged over the scattering angle, the fourth moment after one collision depends on the pairing rule only through the single scalar  $C = \sum_{(a,b)} v_a^2 v_b^2$ . That scalar takes a definite value under any pairing chosen independently of the state, and by a rearrangement argument it is largest when partners have equal speeds and smallest when they are extreme. Any rule whose cross term departs from the exchangeable value therefore moves the fourth moment on the first collision, so the equilibrium is not preserved, for every even particle number  $N \geq 4$ . The identity survives replacing the uniformly resampled scattering angle by any symmetric law shared by all pairs, with the drift attenuated by a computable factor, so small-angle scattering does not evade the conclusion. A law that varies from pair to pair, as a physical cross-section does, falls outside the reduction; Remark 4.5 records why.

The consequence is specific to one way of using the sort. Where the order of the particles is what selects their collision partners, aligning them with a low-discrepancy point set requires ordering by the state, and an ordering that carries velocity information then biases the cross term. Within that adaptation the alignment producing the variance reduction and the bias destroying the equilibrium are the same structure, and no choice of sorting key separates them. Where instead the sort assigns low-discrepancy points to copies of a chain and leaves each copy’s transition law alone, none of this applies; Definition 2.2 draws the line. Section 6 bears this out: the variance reduction is delivered as advertised while the mean squared error is monotonically worsened, with no interior optimum to tune towards.

Two features make the failure difficult to notice, and we think they are worth as much attention as the theorem. The conserved moments are pinned by construction, so a calculation can be wrong by a factor of several hundred in the fourth moment while every routine conservation check reports perfection. And the biased scheme is the more reproducible of the two: its replicates agree to within half a percent of each other while sitting some one hundred and seventy standard deviations

from the truth. Tight agreement between replicates, in this setting, is a symptom rather than a reassurance.

The proof is constructive enough to be useful. The cross term is computable in  $O(N)$  operations from a single state, before any trajectory is run, and comparing it against its exchangeable value tests a proposed scheme in one pass. Section 8 sets out what the test does and does not establish, and identifies the sorting strategies that remain admissible: grouping by a key that carries no velocity information, as the binary-collision schemes in standard use [14] do, leaves the equilibrium intact.

What is established and what is not should be stated plainly. Sections 2 to 4 are proved. Section 5 is proved apart from one hypothesis, displayed as (10) and checked numerically rather than established, and one illustration that is measurement and is marked as such. Section 6 is measured, with the precision of each quantity determined by its sampling behaviour rather than assumed. Section 9 collects what we do not know, of which the most consequential is that the theorem fixes the direction in which the first collision moves the fourth moment and says nothing about where the chain settles. The stationary law of a biased rule remains unidentified.

## 2 The Kac model and its equilibrium

Kac [7] introduced the following caricature of a spatially homogeneous gas. Let  $N$  particles carry scalar velocities  $v = (v_1, \dots, v_N)$ . A collision selects a pair of indices and rotates the two velocities in their coordinate plane, leaving the other components alone. The energy of the pair is conserved exactly, so the total energy is conserved, and after normalising the state moves on the sphere

$$\mathcal{S} = \left\{ v \in \mathbb{R}^N : \sum_i v_i^2 = N \right\}. \quad (1)$$

Throughout,  $N$  is an *even* integer. This is a hypothesis and not a convenience: a collision step matches the  $N$  indices into  $N/2$  disjoint pairs, and a perfect matching exists only for even  $N$ . Every statement below is quantified over even  $N$ .

We consider the step in which all  $N/2$  pairs of a matching collide simultaneously, which is the form a particle code uses. For a pair  $(a, b)$  put  $R^2 = v_a^2 + v_b^2$  and set

$$v_a \leftarrow R \cos \theta, \quad v_b \leftarrow R \sin \theta, \quad (2)$$

with  $\theta$  drawn independently for each pair. Writing  $(v_a, v_b) = R(\cos \phi, \sin \phi)$ , the update (2) replaces  $\phi$  by  $\theta$ ; equivalently, when  $\theta$  is uniform on  $[0, 2\pi)$  it rotates the pair through a uniform angle. Section 4 treats general scattering laws by rotating through an angle  $\theta$  with an arbitrary distribution symmetric about zero, of which the uniform case is one instance.

Let  $\sigma$  denote the uniform measure on  $\mathcal{S}$ . It is the equilibrium of the model, and its low-order structure is available in closed form: the normalised squares  $v_i^2/N$  are marginally  $\text{Beta}(\frac{1}{2}, \frac{N-1}{2})$ , whence

$$\mathbb{E}_\sigma[v_i^4] = \frac{3N}{N+2}, \quad \mathbb{E}_\sigma[Q] = \frac{3N^2}{N+2}, \quad \mathbb{E}_\sigma[Q_6] = \frac{15N^3}{(N+2)(N+4)}, \quad (3)$$

where  $Q(v) = \sum_i v_i^4$  and  $Q_6(v) = \sum_i v_i^6$ . These exact values matter in what follows: they let bias be computed rather than estimated.

**Definition 2.1.** A *pairing rule* assigns to each state a perfect matching of  $\{1, \dots, N\}$ , possibly at random. The rule is *state-independent* if the matching is chosen without reference to  $v$ , whether deterministically or from a distribution not depending on  $v$ , and *state-dependent* otherwise.

The object of this paper is a particular way of bringing Array-RQMC to a collision model, and it should be named, because it is not the only one.

**Definition 2.2.** By a *state-dependent collision pairing rule* we mean the adaptation in which the particles of one system are ordered by their own state and that order is used to choose which particles collide with which. This is to be distinguished from Array-RQMC as ordinarily formulated, in which the objects sorted are independent copies of a Markov chain and the sort decides only which low-discrepancy point each copy receives, leaving the transition law of every copy untouched.

The distinction is the whole of what follows. Sorting copies of a chain is a statement about the assignment of randomness and changes no transition law. Sorting the particles of one system to choose collision partners changes the transition kernel itself, and it is that modified kernel which the next section shows cannot preserve  $\sigma$ . Nothing here rules out an Array-RQMC construction for a Kac system that leaves the collision partners alone.

The distinction is exactly the one that matters, and one direction is immediate.

**Proposition 2.3.** *Under any state-independent pairing rule,  $\sigma$  is invariant.*

*Proof.* The update (2) applied across a matching acts on  $v$  as a product of plane rotations in disjoint coordinate planes. Within the plane of a pair the rotation carries the incoming phase  $\phi$  to  $\theta$ , so the angle turned through is  $\theta - \phi$ , which is not a function of the sampled randomness alone. It is here that uniformity is used: since the uniform law on the circle is invariant under shifts,  $\theta - \phi$  is uniform and independent of  $\phi$ , hence independent of  $v$ . (For the general scattering law of Lemma 4.4 the pair is rotated through  $\theta$  directly and no such step is needed.)

Write the step as  $v \mapsto R(\omega)v$ , where  $\omega$  collects the matching and the turned angles; by the previous paragraph  $\omega$  is independent of  $v$  whenever the matching is. For  $v \sim \sigma$  and any measurable  $E$ , conditioning on  $\omega$  and using the rotation invariance of  $\sigma$  gives  $\Pr(R(\omega)v \in E) = \mathbb{E}_\omega[\sigma(R(\omega)^{-1}E)] = \sigma(E)$ .  $\square$

Proposition 2.3 makes no use of randomness in the matching. A single fixed matching, held for all time, preserves  $\sigma$  as surely as a freshly drawn one, though of course it fails to mix. Neither determinism nor structure in the pairing can therefore be the source of any failure of invariance; only dependence on the state can be.

A state-dependent rule is a piecewise rotation: the sphere partitions into regions on which the matching is constant, and a different rotation is applied on each. The following records when such a map preserves  $\sigma$ . It is not used later, but it locates the mechanism and explains the exceptional case  $N = 2$ , where only one matching exists and no state dependence is possible.

**Proposition 2.4.** *Let  $\{A_i\}$  be a countable measurable partition of  $\mathcal{S}$ , let  $R_i$  be rotations, and let  $T(v) = R_i v$  for  $v \in A_i$ . Then  $T_*\sigma = \sigma$  if and only if  $\{R_i(A_i)\}$  is again a partition of  $\mathcal{S}$  up to  $\sigma$ -null sets.*

*Proof.* For measurable  $E$ , since each  $R_i$  preserves  $\sigma$ ,

$$T_*\sigma(E) = \sum_i \sigma(A_i \cap R_i^{-1}E) = \sum_i \sigma(R_i A_i \cap E) = \int_E m d\sigma, \quad m = \sum_i \mathbf{1}_{R_i A_i}.$$

Because  $\sum_i \sigma(R_i A_i) = \sum_i \sigma(A_i) = 1$ , the multiplicity function satisfies  $\int m d\sigma = 1$  for every such  $T$ . Hence  $T_*\sigma = \sigma$  if and only if  $m = 1$  almost everywhere.  $\square$

### 3 Pairing rules

The rules below are used throughout. All but the first are state-dependent. Ranks are always taken with respect to speed  $|v_i|$ , ties broken arbitrarily; on  $\mathcal{S}$  ties occur with probability zero.

- **Random.** A uniform random perfect matching, drawn afresh each step and independent of the state.
- **Sorted.** Order the particles by  $|v_i|$  and pair adjacent ranks:  $(1, 2), (3, 4), \dots$ . Partners have nearly equal speeds.
- **Extremal.** Order by  $|v_i|$  and pair rank  $i$  with rank  $N + 1 - i$ . Partners have maximally unequal speeds.
- **Block of width  $w$ .** Order by  $|v_i|$ , cut the ordered list into consecutive blocks of  $w$ , and draw a uniform random matching inside each block. Taking  $w = 2$  recovers the sorted rule and  $w = N$  recovers the random rule, so  $w$  interpolates between them while changing nothing else.
- **Cell.** Partition the particles by an auxiliary coordinate carrying no velocity information, such as spatial position, and draw a uniform random matching inside each cell. This is the pairing used by binary-collision schemes of Takizuka–Abe type [14].
- **Curve.** Order the particles by the index of a space-filling curve evaluated at  $(x_i, \Phi(v_i/v_{\text{th}}))$ , where  $x$  is position,  $\Phi$  is the standard normal distribution function and  $v_{\text{th}}$  is the thermal speed, then pair adjacent ranks. This is the rule an Array-RQMC implementation uses, following the Hilbert-curve sort of Gerber and Chopin [3] and the comparisons of L’Ecuyer et al. [10].

The sorted, extremal and block rules are idealisations that isolate one effect. The cell and curve rules are the ones a practitioner would actually write down, and Sections 7 and 8 return to them.

### 4 The obstruction

The argument has three ingredients: a rearrangement lemma about matchings, an identity for the one-step fourth moment, and the observation that the two meet in a single scalar. Figure 1 shows all three on one state.

#### 4.1 A rearrangement lemma for matchings

**Lemma 4.1.** *Let  $x_1 \leq x_2 \leq \dots \leq x_{2n}$  be real numbers and, for a perfect matching  $P$  of  $\{1, \dots, 2n\}$ , put  $S(P) = \sum_{(a,b) \in P} x_a x_b$ . Then  $S$  is maximised by the adjacent matching  $\{(1, 2), (3, 4), \dots, (2n - 1, 2n)\}$  and minimised by the extremal matching  $\{(1, 2n), (2, 2n - 1), \dots, (n, n + 1)\}$ . If the  $x_i$  are distinct and  $n \geq 2$ , each extremiser is unique and every other matching gives a value strictly between the two.*

*Proof.* Let  $P$  maximise  $S$  and suppose 1 is matched to  $j$  and 2 to  $k$  with  $j, k \geq 3$  and  $j \neq k$ . Replacing these two pairs by  $(1, 2)$  and  $(j, k)$  changes  $S$  by

$$(x_1 x_2 + x_j x_k) - (x_1 x_j + x_2 x_k) = (x_2 - x_j)(x_1 - x_k) \geq 0,$$

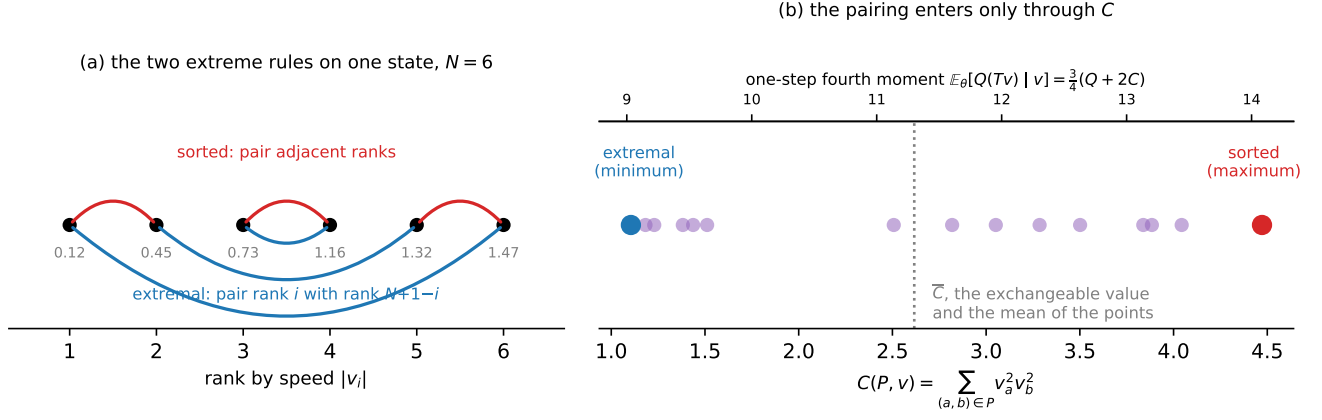


Figure 1: The mechanism, on one state of  $N = 6$  particles drawn from  $\sigma$ . (a) The six speeds in rank order, with the sorted rule pairing adjacent ranks drawn above the axis and the extremal rule, pairing rank  $i$  with rank  $N+1-i$ , drawn below it. (b) All fifteen perfect matchings placed on the  $C$  axis. By Lemma 4.1 the sorted matching gives the largest value and the extremal the smallest; by Lemma 4.6 the exchangeable value  $\bar{C}$  is their mean, marked by the dotted line. The top axis carries the one-step fourth moment that Lemma 4.2 assigns to each, so a rule that systematically selects matchings away from  $\bar{C}$  moves  $\mathbb{E}_\sigma[Q]$  by the corresponding amount. Generated by `kac_fig1_spectrum.v1_0_0.py`.

because  $x_2 \leq x_j$  and  $x_1 \leq x_k$ . Hence some maximiser matches 1 with 2; removing that pair leaves a sorted list of  $2n - 2$  entries and the claim follows by induction. The increment is strictly positive unless  $x_2 = x_j$  or  $x_1 = x_k$ , which gives uniqueness when the  $x_i$  are distinct.

For the minimum, let  $P$  minimise  $S$  and suppose 1 is matched to  $j$  and  $2n$  to  $k$  with  $j, k \notin \{1, 2n\}$ . Replacing these pairs by  $(1, 2n)$  and  $(j, k)$  changes  $S$  by

$$(x_1 x_{2n} + x_j x_k) - (x_1 x_j + x_{2n} x_k) = (x_{2n} - x_j)(x_1 - x_k) \leq 0,$$

because  $x_{2n} \geq x_j$  and  $x_1 \leq x_k$ . Induct as before.  $\square$

For  $n = 2$  the lemma reads  $x_1 x_2 + x_3 x_4 \geq x_1 x_3 + x_2 x_4 \geq x_1 x_4 + x_2 x_3$ , which the two displayed identities verify directly. We note that this is not the classical rearrangement inequality, which concerns two sequences and the sum of their products under a permutation; the statement needed here is about partitioning one sequence into pairs, and the exchange argument above is what supplies it.

## 4.2 The one-step fourth moment

**Lemma 4.2.** *Let  $T$  denote one collision step (2) applied across a matching  $P$ , each angle being uniformly distributed on  $[0, 2\pi)$ . No independence across pairs is required. Then for any  $P$ , state-dependent or not,*

$$\mathbb{E}_\theta[Q(Tv) | v] = \frac{3}{4} \left( Q(v) + 2C(P, v) \right), \quad C(P, v) = \sum_{(a,b) \in P} v_a^2 v_b^2. \quad (4)$$

*Proof.* A pair contributes  $R^4(\cos^4 \theta + \sin^4 \theta)$  to  $Q(Tv)$ , and  $\mathbb{E}[\cos^4 \theta] = \mathbb{E}[\sin^4 \theta] = 3/8$ , so its conditional expectation is  $\frac{3}{4}R^4$ . Summing over pairs and expanding,  $\sum_P (v_a^2 + v_b^2)^2 = \sum_i v_i^4 + 2 \sum_P v_a^2 v_b^2$ .  $\square$

Identity (4) is the engine of everything below, and the point to notice is what it does not contain. The pairing rule enters the one-step fourth moment only through the single scalar  $C$ ; every other quantity in (4) depends on the state alone. Whatever elaborate structure a sorting scheme imposes, its entire effect on this moment is summarised by one number.

The restriction to uniformly resampled angles is inessential.

**Remark 4.3.** Only the marginal law of each angle is used. Writing  $Q(Tv)$  as a sum over the pairs of  $P$  and taking expectations term by term needs linearity alone, so (4) holds whatever the joint law of the family  $(\theta_{ab})_{(a,b) \in P}$ , provided each  $\theta_{ab}$  is uniform. That matters in the present setting: the angles supplied by a randomized quasi-Monte Carlo point set are individually uniform and deliberately correlated, and the lemma covers them.

**Lemma 4.4.** *Suppose instead that the collision rotates the pair through an angle  $\theta$  drawn from one distribution, symmetric about zero and shared by every pair, so that  $(v_a, v_b) = R(\cos \phi, \sin \phi) \mapsto R(\cos(\phi + \theta), \sin(\phi + \theta))$ , and set  $\kappa = \mathbb{E}[\cos 4\theta]$ . Then*

$$\mathbb{E}_\theta[Q(Tv) \mid v] = \kappa Q(v) + (1 - \kappa) \frac{3}{4} (Q(v) + 2C(P, v)). \quad (5)$$

*Proof.* For a single pair,  $v_a^4 + v_b^4 = R^4(1 - \frac{1}{2} \sin^2 2\phi) = R^4(\frac{3}{4} + \frac{1}{4} \cos 4\phi)$ , so that  $R^4 \cos 4\phi = 4(v_a^4 + v_b^4) - 3R^4$ . After the rotation the same expression holds with  $\phi$  replaced by  $\phi + \theta$ , and averaging over  $\theta$  symmetric about zero annihilates the term in  $\sin 4\theta$  and multiplies the term in  $\cos 4\phi$  by  $\kappa$ . Hence

$$\mathbb{E}_\theta[v_a'^4 + v_b'^4] = \frac{3}{4} R^4 + \frac{\kappa}{4} R^4 \cos 4\phi = (1 - \kappa) \frac{3}{4} R^4 + \kappa (v_a^4 + v_b^4).$$

Summing over pairs and using  $\sum_P R^4 = Q + 2C$  gives (5).  $\square$

Uniform resampling is the case  $\kappa = 0$ , where (5) reduces to (4), and  $\kappa = 1$  is the degenerate case in which the collision cannot change  $Q$ . For every  $\kappa < 1$  the pairing still enters only through  $C$ , and the coefficient multiplying it is simply scaled by  $1 - \kappa$ . Small-angle scattering, for which  $\kappa$  is close to one, therefore attenuates the effect described below without removing it.

### 4.3 The exchangeable value

**Remark 4.5.** A common law across pairs is needed, and this is the boundary of the single-scalar reduction. If the angle law varies with the pair, so that  $\kappa_{ab} = \mathbb{E}[\cos 4\theta_{ab}]$  depends on  $(a, b)$ , the same calculation gives

$$\mathbb{E}_\theta[Q(Tv) \mid v] = \sum_{(a,b) \in P} \left( \kappa_{ab} (v_a^4 + v_b^4) + (1 - \kappa_{ab}) \frac{3}{4} R_{ab}^4 \right), \quad R_{ab}^2 = v_a^2 + v_b^2,$$

which depends on which  $\kappa$  is attached to which pair and so is not a function of  $Q$  and  $C$  alone; states agreeing in  $(Q, C)$  but differing in this quantity exist. The reduction to a single scalar, and with it Theorem 4.8, is therefore a statement about a common scattering law. A physical cross-section, whose angle law depends on the relative speed of the colliding pair, lies outside it.

**Lemma 4.6.** *For an exchangeable rule, meaning one that draws the matching uniformly at random independently of the state, the conditional expectation of  $C$  given  $v$  is*

$$\overline{C}(v) = \frac{N^2 - Q(v)}{2(N - 1)}. \quad (6)$$

Substituting (6) into (4) and taking expectations over  $\sigma$  gives an affine map on  $m = \mathbb{E}_\sigma[Q]$ ,

$$\Phi(m) = \frac{3}{4} \cdot \frac{m(N-2) + N^2}{N-1}, \quad (7)$$

whose slope  $\frac{3}{4}(N-2)/(N-1)$  is less than one for every  $N$ , and whose unique fixed point is  $m^* = 3N^2/(N+2) = \mathbb{E}_\sigma[Q]$ . For any state-independent rule, exchangeable or not,  $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\bar{C}]$ .

*Proof.* By exchangeability,  $\mathbb{E}[C | v] = (N/2) \mathbb{E}[v_a^2 v_b^2]$  for a uniformly chosen pair  $(a, b)$ , which equals  $(2(N-1))^{-1} \sum_{i \neq j} v_i^2 v_j^2$ . Using  $\sum_i v_i^2 = N$  this is  $(N^2 - Q)/(2(N-1))$ , giving (6), and substitution yields (7). Solving  $\Phi(m) = m$  gives  $m(N+2) = 3N^2$ . The identification of  $m^*$  with  $\mathbb{E}_\sigma[Q]$  is (3).

For the last claim, let  $P_0$  be any fixed matching. Since  $\sigma$  is invariant under permutations of the coordinates,  $\mathbb{E}_\sigma[v_a^2 v_b^2]$  takes the same value for every pair  $a \neq b$ , so summing over the  $N/2$  pairs of  $P_0$  gives  $\mathbb{E}_\sigma[C(P_0, \cdot)] = (N/2) \mathbb{E}_\sigma[v_1^2 v_2^2] = (N^2 - \mathbb{E}_\sigma[Q])/(2(N-1)) = \mathbb{E}_\sigma[\bar{C}]$ . A state-independent rule is a mixture of fixed matchings, so the same holds for it.  $\square$

**Remark 4.7.** The distinction matters. A fixed matching is state-independent by Definition 2.1, but for it  $\mathbb{E}[C | v] = C(P_0, v)$ , which by Lemma 4.1 may be the largest or the smallest cross term available at that state rather than the average. What survives for every state-independent rule is the equality of  $\sigma$ -expectations, and that is all Theorem 4.8 uses.

Two remarks. First,  $Q$  is a random variable and the fixed point in (7) is of the induced map on its expectation, not of a scalar recursion; the map closes on the mean because it is affine in  $Q$ , which is what makes the argument legitimate. Second, Lemma 4.6 is consistent with Proposition 2.3, as it must be: a state-independent rule leaves  $\mathbb{E}_\sigma[Q]$  where it found it.

## 4.4 The theorem

**Theorem 4.8.** *Let  $P$  be a pairing rule for which  $\mathbb{E}_\sigma[C(P(v), v)] \neq \mathbb{E}_\sigma[\bar{C}(v)]$ . Then  $\sigma$  is not invariant. If the left-hand side is the larger, one step from  $\sigma$  strictly increases  $\mathbb{E}_\sigma[Q]$ ; if it is the smaller, the step strictly decreases it. Under a scattering law with parameter  $\kappa < 1$  the same conclusion holds with the change in  $\mathbb{E}_\sigma[Q]$  scaled by  $1 - \kappa$ .*

*Proof.* Integrating (4) against  $\sigma$  gives  $\mathbb{E}_\sigma[Q(Tv)] = \frac{3}{4}(\mathbb{E}_\sigma[Q] + 2\mathbb{E}_\sigma[C])$ . By Lemma 4.6, replacing  $\mathbb{E}_\sigma[C]$  by  $\mathbb{E}_\sigma[\bar{C}]$  returns  $\mathbb{E}_\sigma[Q]$  exactly. The right-hand side is strictly increasing in  $\mathbb{E}_\sigma[C]$ , so  $\mathbb{E}_\sigma[C] \neq \mathbb{E}_\sigma[\bar{C}]$  forces  $\mathbb{E}_\sigma[Q(Tv)] \neq \mathbb{E}_\sigma[Q]$  with the stated sign. Were  $\sigma$  invariant the two would agree. The final claim follows the same way from (5), whose dependence on  $C$  carries the factor  $1 - \kappa$ .  $\square$

The hypothesis is an inequality between integrals, which is weaker than pointwise domination and is all the proof uses. Two ratios follow from it and must be kept apart for the rest of the paper. Write

$$r(v) = \frac{C(P(v), v)}{\bar{C}(v)} \quad \text{and} \quad r_\sigma = \frac{\mathbb{E}_\sigma[C]}{\mathbb{E}_\sigma[\bar{C}]},$$

the first read from a single state and the second an average over the equilibrium law. Theorem 4.8 constrains  $r_\sigma$ . What a practitioner can compute from a state in hand is  $r(v)$ , and the two are not interchangeable: a fixed matching is state-independent, so  $r_\sigma = 1$  exactly by Lemma 4.6, while  $r(v)$  at a given state may be the largest or the smallest value available there.

**Corollary 4.9.** *For even  $N \geq 4$ , the measure  $\sigma$  is invariant under neither the sorted nor the extremal rule. Sorted pairing strictly increases  $\mathbb{E}_\sigma[Q]$  on the first step, and extremal pairing strictly decreases it. For  $N = 2$  there is a unique matching, no rule can depend on the state, and  $\sigma$  is preserved.*

*Proof.* Apply Lemma 4.1 with  $x_i = v_i^2$ . The sorted rule realises the maximum of  $C$  over matchings pointwise and the extremal rule the minimum. Under  $\sigma$  the  $v_i^2$  are almost surely distinct and, for  $N \geq 4$ , at least three matchings exist, so by the uniqueness clause the maximum strictly exceeds the average over matchings almost surely, and likewise the minimum falls strictly below it. Taking expectations gives  $\mathbb{E}_\sigma[C] \neq \mathbb{E}_\sigma[\overline{C}]$  with the stated sign in each case, and Theorem 4.8 applies. The case  $N = 2$  is Proposition 2.3.  $\square$

The two directions describe different failures. Pairing partners of equal speed drives the fourth moment up, which under a fixed second moment means mass accumulating near zero while a thin population carries the energy. Pairing extremes drives it down, producing a distribution with the tails removed. Neither is the Maxwellian, and Section 6 exhibits both.

## 5 Necessity without sufficiency

Theorem 4.8 gives a condition whose violation certifies that  $\sigma$  is not invariant. It is natural to hope that its satisfaction certifies the opposite, so that a scheme could be cleared by a single inexpensive test. It does not, and the reason is that the fourth moment is the first member of a family whose later members carry independent information.

**Lemma 5.1.** *With  $Q_6(v) = \sum_i v_i^6$ , and angles drawn independently and uniformly,*

$$\mathbb{E}_\theta[Q_6(Tv) \mid v] = \frac{5}{8} \left( Q_6(v) + 3 D(P, v) \right), \quad D(P, v) = \sum_{(a,b) \in P} v_a^2 v_b^2 (v_a^2 + v_b^2).$$

*For an exchangeable rule the conditional expectation of  $D$  given  $v$  is  $\overline{D}(v) = (N Q(v) - Q_6(v)) / (N - 1)$ .*

*Proof.* A pair contributes  $R^6(\cos^6 \theta + \sin^6 \theta)$ , and  $\mathbb{E}[\cos^6 \theta] = \binom{6}{3} 2^{-6} = 5/16$ , so the conditional expectation of the contribution is  $\frac{5}{8} R^6$ . Expanding  $R^6 = v_a^6 + v_b^6 + 3v_a^2 v_b^2 (v_a^2 + v_b^2)$  and summing over pairs gives the identity. For  $\overline{D}$ , exchangeability gives  $\mathbb{E}[D \mid v] = (2(N - 1))^{-1} \sum_{i \neq j} (v_i^4 v_j^2 + v_i^2 v_j^4) = (N - 1)^{-1} \sum_{i \neq j} v_i^4 v_j^2$ , and  $\sum_i v_i^2 = N$  turns this into  $(N Q - Q_6) / (N - 1)$ .  $\square$

Substituting  $\overline{D}$  into Lemma 5.1 and using (3) returns  $\mathbb{E}_\sigma[Q_6]$ , as Proposition 2.3 requires. For  $N \geq 6$  the functional  $D$  is not a function of  $C$ , so the two conditions are logically independent, and there is no reason to expect one to imply the other. At  $N = 4$  they are not independent at all.

**Proposition 5.2** (collapse at four particles). *Let  $N = 4$ . Then for every  $v \in \mathcal{S}$  and every matching  $P$ ,*

$$D(P, v) - \overline{D}(v) = N(C(P, v) - \overline{C}(v)). \quad (8)$$

*Consequently a pairing rule satisfies  $\mathbb{E}_\sigma[D] = \mathbb{E}_\sigma[\overline{D}]$  if and only if it satisfies  $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\overline{C}]$ : at four particles the sixth-moment condition carries no information beyond the fourth.*

*Proof.* Write  $x_i = v_i^2$ . Let  $P$  and  $P'$  be matchings agreeing except on four indices, with  $(a, b), (c, d) \in P$  and  $(a, c), (b, d) \in P'$ . The common terms cancel, leaving

$$C(P) - C(P') = x_a x_b + x_c x_d - x_a x_c - x_b x_d = (x_a - x_d)(x_b - x_c).$$

Collecting the same four terms of  $D$ ,

$$\begin{aligned} D(P) - D(P') &= x_a x_b (x_a + x_b) + x_c x_d (x_c + x_d) - x_a x_c (x_a + x_c) - x_b x_d (x_b + x_d) \\ &= (x_b - x_c) [(x_a^2 - x_d^2) + (x_b + x_c)(x_a - x_d)] \\ &= (x_a - x_d)(x_b - x_c)(x_a + x_b + x_c + x_d), \end{aligned}$$

so that

$$D(P) - D(P') = (C(P) - C(P'))(x_a + x_b + x_c + x_d). \quad (9)$$

At  $N = 4$  there are exactly three matchings, any two of them differ by such an exchange, and the four indices it touches are all of them, so the multiplier in (9) is  $\sum_i x_i = N$ . Hence  $D(P) - N C(P)$  takes one value at all three matchings. Averaging that constant over the three returns  $\bar{D} - N \bar{C}$ , and subtracting gives (8). Since (8) holds for every matching it holds for the matching a rule produces, and taking expectations under  $\sigma$  gives  $\mathbb{E}_\sigma[D] - \mathbb{E}_\sigma[\bar{D}] = N(\mathbb{E}_\sigma[C] - \mathbb{E}_\sigma[\bar{C}])$ , from which the equivalence is immediate.  $\square$

The multiplier in (9) is the sum over the four indices the exchange touches, not over the whole system, so the collapse is special to  $N = 4$ : for  $N \geq 6$  that sum varies from one exchange to the next and  $D$  is no longer an affine function of  $C$  across matchings. The hierarchy of this section therefore begins at  $N = 6$ .

For  $N \geq 6$  the fourth-moment condition is not expected to imply the rest, and under a condition recorded in the proposition below, and verified numerically rather than proved, the failure can be exhibited exactly.

**Proposition 5.3** (the converse of Theorem 4.8 fails). *Let  $N \geq 6$  be even. For a pairing rule write  $\delta_C$  for  $\mathbb{E}_\sigma[C] - \mathbb{E}_\sigma[\bar{C}]$  and  $\delta_D$  for  $\mathbb{E}_\sigma[D] - \mathbb{E}_\sigma[\bar{D}]$ , and let  $\delta_C^{\text{srt}}, \delta_D^{\text{srt}}$  and  $\delta_C^{\text{ext}}, \delta_D^{\text{ext}}$  denote their values for the sorted and extremal rules. If*

$$\delta_C^{\text{srt}} \delta_D^{\text{ext}} \neq \delta_C^{\text{ext}} \delta_D^{\text{srt}}, \quad (10)$$

*then there is a state-dependent pairing rule with  $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\bar{C}]$  exactly, so that the hypothesis of Theorem 4.8 is not met, and for which  $\sigma$  is nonetheless not invariant.*

*Proof.* Let the rule apply the sorted rule with probability  $p$  and the extremal rule with probability  $1 - p$ , the choice drawn independently of the state. Both  $\delta_C$  and  $\delta_D$  are expectations, so they are affine in  $p$ :  $\delta_C(p) = p \delta_C^{\text{srt}} + (1 - p) \delta_C^{\text{ext}}$ , and likewise for  $\delta_D$ . By Corollary 4.9,  $\delta_C^{\text{srt}} > 0 > \delta_C^{\text{ext}}$ , so

$$p^* = \frac{-\delta_C^{\text{ext}}}{\delta_C^{\text{srt}} - \delta_C^{\text{ext}}} \in (0, 1) \quad \text{and} \quad \delta_C(p^*) = 0$$

exactly, not merely to within a sampling resolution. At that  $p^*$ ,

$$\delta_D(p^*) = \frac{\delta_C^{\text{srt}} \delta_D^{\text{ext}} - \delta_C^{\text{ext}} \delta_D^{\text{srt}}}{\delta_C^{\text{srt}} - \delta_C^{\text{ext}}},$$

whose numerator is nonzero precisely under (10). Now integrate Lemma 5.1 against  $\sigma$ , which gives  $\mathbb{E}_\sigma[Q_6(Tv)] = \frac{5}{8}(\mathbb{E}_\sigma[Q_6] + 3\mathbb{E}_\sigma[D])$ , and replacing  $\mathbb{E}_\sigma[D]$  by  $\mathbb{E}_\sigma[\bar{D}]$  there returns  $\mathbb{E}_\sigma[Q_6]$ . Hence  $\delta_D(p^*) \neq 0$  forces  $\mathbb{E}_\sigma[Q_6(Tv)] \neq \mathbb{E}_\sigma[Q_6]$ , and  $\sigma$  is not invariant.  $\square$

Condition (10) asks that the sorted and extremal rules move the fourth and sixth moments by non-proportional amounts. It fails at  $N = 4$ , where Proposition 5.2 makes the two sides equal, which is why the proposition is stated for  $N \geq 6$ . We have checked it numerically at  $N = 6, 8, 16$  and  $64$ , where the ratio  $\delta_C^{\text{ext}} \delta_D^{\text{srt}} / (\delta_C^{\text{srt}} \delta_D^{\text{ext}})$  is 1.12, 1.21, 1.38 and 1.62, moving further from unity as  $N$  grows; the resulting one-step change in  $\mathbb{E}_\sigma[Q_6]$  runs from about two percent at  $N = 6$  to fifteen percent at  $N = 64$ . A proof of (10) for general  $N$  we do not have.

Proposition 5.3 defeats the condition of Theorem 4.8, which is an equality of  $\sigma$ -expectations. It does not defeat the test as a practitioner applies it, on a single state: the rule takes the sorted branch or the extremal one, so at  $N = 64$  it reads  $C/\overline{C}$  near 2.9 or near 0.14 and never near unity. Defeating the one-state reading is a separate requirement, and the construction below meets it.

**Remark 5.4** (the one-state reading can be passed by a bad rule). Proposition 5.3 defeats an equality of  $\sigma$ -expectations. The test as a practitioner applies it reads  $C/\overline{C}$  from a single state, and that reading too can sit near unity for a rule that does not preserve  $\sigma$ . The evidence for this is numerical, and is set out below rather than proved.

Select a fraction  $f$  of the particles uniformly at random, pair those among themselves by the sorted rule, and pair the remainder among themselves by the extremal rule. At  $f = 1$  the rule is sorted and  $\mathbb{E}_\sigma[C] > \mathbb{E}_\sigma[\overline{C}]$ ; at  $f = 0$  it is extremal and  $\mathbb{E}_\sigma[C] < \mathbb{E}_\sigma[\overline{C}]$ , both by Corollary 4.9. As  $f$  increases through its admissible values the difference  $\mathbb{E}_\sigma[C] - \mathbb{E}_\sigma[\overline{C}]$  changes sign in steps of order  $N^{-1}$ , so some  $f^*$  makes it vanish to that order. Unlike the mixture of Proposition 5.3, this rule applies a single matching at every state, so its one-state reading of  $C/\overline{C}$  sits near unity instead of alternating between two values far from it. The two sub-populations distort in opposite directions and their contributions to  $C$  cancel, while nothing forces their contributions to any other functional to cancel with them. That last step is an expectation rather than an argument, and what follows is measurement.

Locating  $f^*$  precisely is not possible and not necessary. The cross term is estimated from a finite sample and varies by a few percent between draws, so the crossing is determined only to within that resolution, and quantities read off at an interpolated  $f^*$  inherit the imprecision, the more so the higher the moment. Five independent determinations at  $N = 2048$  placed  $f^*$  between 0.297 and 0.308, returned  $C/\overline{C}$  between 0.94 and 1.03, and returned  $D/\overline{D}$  between 1.4 and 1.6; the stationary sixth moment, a sixth moment of a heavy-tailed law read off at an interpolated crossing, ranged from 2.0 to 3.7 across the same five. The claim of Remark 5.4 is an existence claim, and the reproducible form of it is this. At  $N = 2048$  there are blend fractions near 0.30 at which  $C/\overline{C}$  agrees with unity to within the resolution of the diagnostic itself, while  $D/\overline{D}$  is about 1.5, the stationary sixth moment is several times the Maxwellian value, and only about half the mass beyond two thermal speeds survives. A test on  $C$  alone would clear such a rule.

The pair  $\{C/\overline{C}, D/\overline{D}\}$  separates every rule we have examined, including this one. Whether some finite collection of such conditions is sufficient, or whether each even moment contributes an independent constraint indefinitely, we do not know; the question is stated as Conjecture 9.2 below.

## 6 Numerical study

Four experiments follow, each answering one question. All were run through a generator that tags every reported quantity with its sampling class, computes a standard error, repeats the whole experiment under two independent seeds and reports only the digits agreeing between them. Where a quantity supports one significant figure, one is given.

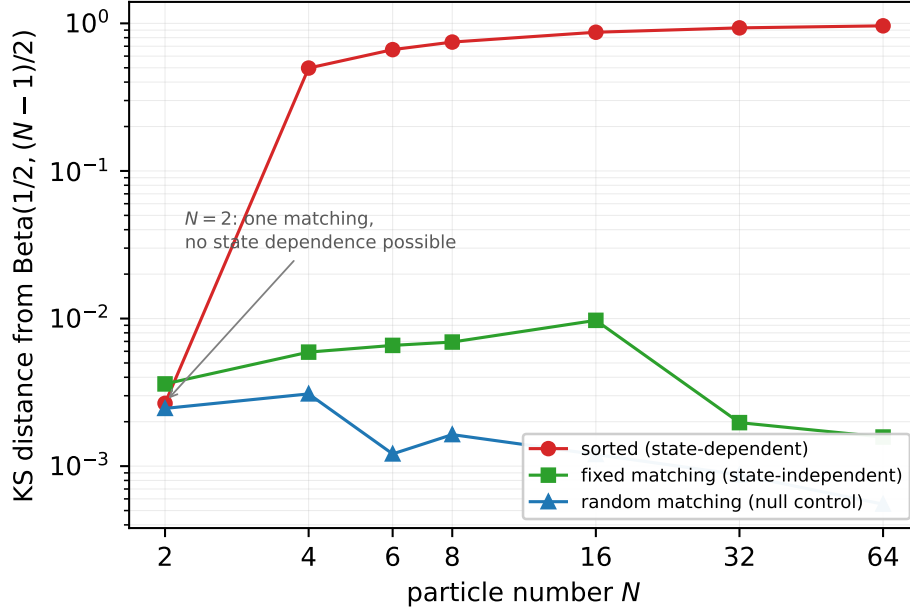


Figure 2: Kolmogorov–Smirnov distance between the empirical law of  $v_i^2/N$  and its exact uniform-on-sphere marginal  $\text{Beta}(\frac{1}{2}, \frac{N-1}{2})$ , after 3000 burn-in steps and 60 collections at intervals of 25, over 400 chains. Random matching is the null control. A fixed state-independent matching sits modestly above it, by a factor between one and a half and eight, which slow mixing accounts for and Proposition 2.3 forbids attributing to bias; the departure of the sorted rule is two to three orders of magnitude larger, so neither determinism nor structure in the pairing breaks the equilibrium. The sorted rule matches the control at  $N = 2$ , where only one matching exists and no rule can depend on the state, and departs from it at every even  $N \geq 4$ , as Corollary 4.9 requires. Generated by `kac_manuscript_data_v1.0_0.py`.

## 6.1 Does the one-step identity hold in the code?

Everything downstream rests on Lemma 4.2, which is an identity about one collision step. It is worth testing directly rather than inferring it from agreement further along. We draw  $v \sim \sigma$ , apply a single step under each pairing rule, and compare the measured  $\mathbb{E}_\sigma[Q(Tv)]$  against the exact prediction  $\frac{3}{4}(Q + 2C)$  evaluated on the same states.

Table 1 reports the result. The identity holds across rules whose cross-term ratios differ by a factor of twenty-four, with a largest relative discrepancy of 0.028% and no  $|d/\text{SE}|$  above one. The two state-independent rules, random and fixed, return  $r(v)$  at unity as Lemma 4.6 requires, which is a check on the pairing code independent of the quantity being tested.

The curve rule is not in the table because it is defined on (position, velocity) pairs and a draw from  $\sigma$  carries no positions. Tested instead on the loads of Section 7, where  $r(v) = 2.713$  at  $N = 2048$  and 2.834 at  $N = 8192$ , it agrees with Lemma 4.2 to 0.008% and 0.005% respectively. Lemma 4.2 is a statement conditional on  $v$ , so it applies to those states as it does to a draw from  $\sigma$ ; what it describes is the collision step alone, and Section 7 composes that step with the projection recorded there.

Table 1: One-step test of Lemma 4.2 at  $N = 2048$ , 4000 states per rule, angles uniform. The rules span the cross-term ratio from 0.12 to 2.99. Every discrepancy is under 0.03% and within one standard error of zero; a second independent seed agrees. The  $w = 512$  row is given to three decimals because two would fall on a rounding tie. Generated by `kac_onestep_test_v1.0.0.py`.

| rule             | $r(v)$ | predicted | measured | rel. diff. | $d/SE$ |
|------------------|--------|-----------|----------|------------|--------|
| random           | 1.002  | 6137.15   | 6138.89  | +0.028%    | +1.00  |
| fixed            | 0.999  | 6142.69   | 6142.70  | +0.000%    | +0.01  |
| sorted           | 2.993  | 9204.53   | 9202.25  | -0.025%    | -0.64  |
| extremal         | 0.122  | 4792.46   | 4793.40  | +0.020%    | +0.73  |
| block, $w = 64$  | 2.889  | 9036.08   | 9038.37  | +0.025%    | +0.71  |
| block, $w = 512$ | 2.279  | 8098.275  | 8099.592 | +0.016%    | +0.54  |

Table 2: Moment ladder across the block width at  $N = 2048$ ,  $t = 40$ , eight replicates under each of two seeds, normalised so the Maxwellian value is unity. The fourth and sixth moments are heavy-tailed at small  $w$  and support one significant figure; the second moment is exact and is given in full. The  $C/\overline{C}$  column averages over twelve independently drawn equilibrium states, so it estimates the ensemble value; the same quantity in Table 4 is read from one fixed load. Across 600 states the sorted rule gives 2.99 with a load-to-load standard deviation of 0.11, which is the spread a practitioner comparing a single reading against unity must allow for.

| $w$ | $C/\overline{C}$ | $m_2 = \langle v^2 \rangle$ | $m_4$           | $m_6$           | tail               |
|-----|------------------|-----------------------------|-----------------|-----------------|--------------------|
| 2   | 3.0              | 1.0000000000000000          | $5 \times 10^2$ | $2 \times 10^5$ | $3 \times 10^{-2}$ |
| 32  | 3.0              | 1.0000000000000000          | $6 \times 10^1$ | $3 \times 10^3$ | $3 \times 10^{-1}$ |
| 512 | 2.3              | 1.0000000000000000          | $4 \times 10^0$ | $2 \times 10^1$ | $2 \times 10^0$    |
| $N$ | 1                | 1.0000000000000000          | $1 \times 10^0$ | $1 \times 10^0$ | $1.0 \times 10^0$  |

## 6.2 Is the failure finite in $N$ ?

Figure 2 answers this. The measure  $\sigma$  is preserved at  $N = 2$ , where only one matching exists and no rule can depend on the state, and is not preserved at any even  $N \geq 4$ , with the departure growing to a Kolmogorov–Smirnov distance of 0.96 by  $N = 64$ . A fixed, deterministic, state-independent matching behaves quite differently. Its Kolmogorov–Smirnov distance rises from 0.0036 at  $N = 2$  to 0.0097 at  $N = 16$  and falls back to 0.0016 by  $N = 64$ , sitting between one and a half and eight times the random-matching arm rather than on top of it. That residual elevation is what slow mixing produces: Proposition 2.3 guarantees that such a rule preserves  $\sigma$  exactly, but a chain that never rearranges its pairs approaches its invariant law slowly, so a finite sample still shows a gap. The separation that matters is of a different order altogether. At  $N = 64$  the sorted rule stands at 0.96 against the fixed rule’s 0.0016, a factor of six hundred. Neither determinism nor structure in the pairing carries the damage; only dependence on the state does. This is Corollary 4.9 seen in the stationary law rather than in one step.

## 6.3 Which observables carry the damage?

Table 2 locates it. A point of method belongs here, because it decides what the table is evidence of. After each collision sweep the code applies a projection: it removes the global mean velocity

and then rescales to restore the kinetic energy,

$$v \mapsto (v - \bar{v} \mathbf{1}) \sqrt{\mathcal{E}_0/\mathcal{E}}, \quad \mathcal{E} = \frac{1}{2} \|v - \bar{v} \mathbf{1}\|^2,$$

with  $\mathcal{E}_0$  the initial kinetic energy. The chain actually simulated is therefore the composition of this map with the collision step, not the collision step alone.

Which results this covers should be stated, because it is not all of them. Table 2, the moment ladder of Section 6, the stationary quantities quoted for the construction of Section 5, and the applied instance of Section 7 all run the projected chain. The finite- $N$  sweep of Figure 2 does not: there the rotation conserves the energy of each pair exactly, the state stays on  $\mathcal{S}$  without help, and no projection is applied at any step. The one-step test of Table 1 likewise uses the collision step alone, which is what Lemma 4.2 describes. The second moment is exact to fifteen digits at every block width, including the most damaged, but for two reasons that should be kept apart. The collision conserves the energy of each pair, so the total energy is conserved by the dynamics whatever the pairing rule does, and the projection then enforces it against rounding. The momentum is different: the collision does not conserve it, and it is pinned by the projection alone. In neither case is the agreement evidence about the pairing rule.

Two things follow. The first is that the projection cannot be what produces the damage recorded in Table 2. Sampling the equilibrium of the projected chain directly and comparing with  $\sigma$  gives a shift in  $\mathbb{E}_\sigma[Q]$  of  $-0.04\%$  at  $N = 64$  and  $-0.0007\%$  at  $N = 2048$ , diminishing with  $N$  and smaller than the fourth-moment error by four orders of magnitude. The second is that the agreement of the conserved moments is not independent evidence of anything, since the projection enforces exactly those moments. Meanwhile the fourth moment is wrong by a factor of order  $10^2$  and the sixth by  $10^5$ , and the mass beyond two thermal speeds falls to a few percent of its Maxwellian value.

The consequence for practice is the important part of this section. **A calculation can be wrong in this way while every conservation check it performs reports perfection.** The quantities a particle code monitors as a matter of routine are exactly the ones the projection protects.

## 6.4 Can it be repaired?

If the low-discrepancy angles supplied the variance reduction and the sorting supplied the bias, one could keep the first and discard the second. Table 3 shows that this fails. Low-discrepancy angles with random pairing leave the variance of  $\langle v^4 \rangle$  where Monte Carlo had it, the two agreeing within their own sampling error. Per-step stratification is washed out by the mixing of the dynamics over 800 steps. What survives the mixing is the sort, which re-imposes low-discrepancy structure on the state at every step, and the sort is what biases the equilibrium.

Table 4 and Figure 3(a) show why no intermediate setting rescues the method either. The mean squared error is monotone in the block width for both observables tested: any amount of velocity-correlated alignment is net harmful, because the bias it buys always outruns the variance it saves. There is no interior optimum to tune towards.

The variance reduction itself is real but observable-dependent, and is reported here per observable rather than in general. On  $\langle |v| \rangle$  full alignment reduces the variance against random pairing by something of order  $10^3$ , the order Array-RQMC promises; the factor is not reproducible beyond that, being a ratio of two small variance estimates. On  $\langle v^4 \rangle$  there is no reduction against Monte Carlo at all. The obstruction rests on the monotonicity, which holds on both observables, and not on the size of any reduction.

Table 3: The candidate fix.  $N = 2048$ , 800 steps, 100 randomizations, initial ensemble held fixed. Bias is against the exact equilibrium  $\mathbb{E}_\sigma[\langle v^4 \rangle] = 3N/(N+2) = 2.997073$ . Biases and MSEs to three significant figures, variances to two.

| scheme                       | bias     | variance | MSE     |
|------------------------------|----------|----------|---------|
| MC (random, iid)             | 3.03e−3  | 1.2e−2   | 1.18e−2 |
| RQMC (random, stratified)    | −3.86e−3 | 1.0e−2   | 1.02e−2 |
| A-iid (sorted, iid)          | 1.50e+3  | 1.2e+5   | 2.36e+6 |
| Array-RQMC (sorted, aligned) | 2.01e+3  | 1.2e−2   | 4.06e+6 |

Table 4: Trade-off across the block width  $w$ , aligned angles throughout, 60 randomizations.  $w = 2$  is the sorted rule and  $w = N$  the random rule.  $C/\overline{C}$  here is read from the single fixed load, not averaged over states, so it carries the full load-to-load spread quoted in Table 2.

| $w$  | $C/\overline{C}$ | $\text{bias}\langle v^4 \rangle$ | $\text{MSE}\langle v^4 \rangle$ | $\text{bias}\langle  v  \rangle$ | $\text{var}\langle  v  \rangle$ | $\text{MSE}\langle  v  \rangle$ |
|------|------------------|----------------------------------|---------------------------------|----------------------------------|---------------------------------|---------------------------------|
| 2    | 2.94             | 2.01e+3                          | 4.06e+6                         | −7.21e−1                         | 3.6e−9                          | 5.20e−1                         |
| 8    | 2.93             | 2.00e+3                          | 3.99e+6                         | −7.17e−1                         | 1.3e−6                          | 5.15e−1                         |
| 64   | 2.84             | 7.03e+2                          | 5.34e+5                         | −6.10e−1                         | 1.4e−4                          | 3.72e−1                         |
| 512  | 2.26             | 9.05e+0                          | 8.24e+1                         | −3.91e−1                         | 1.1e−5                          | 1.53e−1                         |
| 2048 | 0.991            | 1.57e−2                          | 1.12e−2                         | −1.80e−4                         | 2.0e−5                          | 1.99e−5                         |

## 6.5 Does the diagnostic work?

Figure 3(b) shows that  $r(v)$  does its job: it tracks the measured bias monotonically and costs  $O(N)$  operations on one state, before any trajectory is run. But it is an estimate of  $r_\sigma$  and carries the load-to-load spread of Table 2, about 0.11 at  $N = 2048$ ; differences below roughly ten percent should not be read as real. The inference also runs one way only. A value of  $r(v)$  far from unity is evidence that  $r_\sigma \neq 1$  and so condemns a scheme; a value near unity clears it only of the fourth-moment obstruction, by Proposition 5.3, and does not by itself establish  $r_\sigma = 1$ .

## 7 The applied instance

The problem that prompted this work is a one-dimensional collisional particle-in-cell test: charged particles on a periodic domain with a grid-based field solve and a binary collision operator of the kind treated above, run to a warm equilibrium and compared against it through a kernel discrepancy between the empirical joint law of  $(x, v)$  and the target. The implementation follows Array-RQMC practice, ordering the particles by the index of a Hilbert curve evaluated at  $(x, \Phi(v/v_{\text{th}}))$  and pairing adjacent ranks, which is the curve rule of Section 3.

One thing this test does not do should be said at the outset. Its collision operator is the Kac rotation of Section 2, with the scattering angle drawn uniformly and every selected pair colliding; it is not a physical cross section. The section therefore shows the obstruction reaching a production-shaped code with a grid, a field solve and a real particle load, and it does not show the obstruction escaping the caricature. Those are different claims and only the first is made here.

That rule is not the sorted rule of Corollary 4.9, so the corollary does not apply to it directly. Theorem 4.8 does, because its hypothesis is a computable property of a rule at a state. Table 5 evaluates it on the implementation’s own Maxwellian load: the cross term exceeds its exchangeable

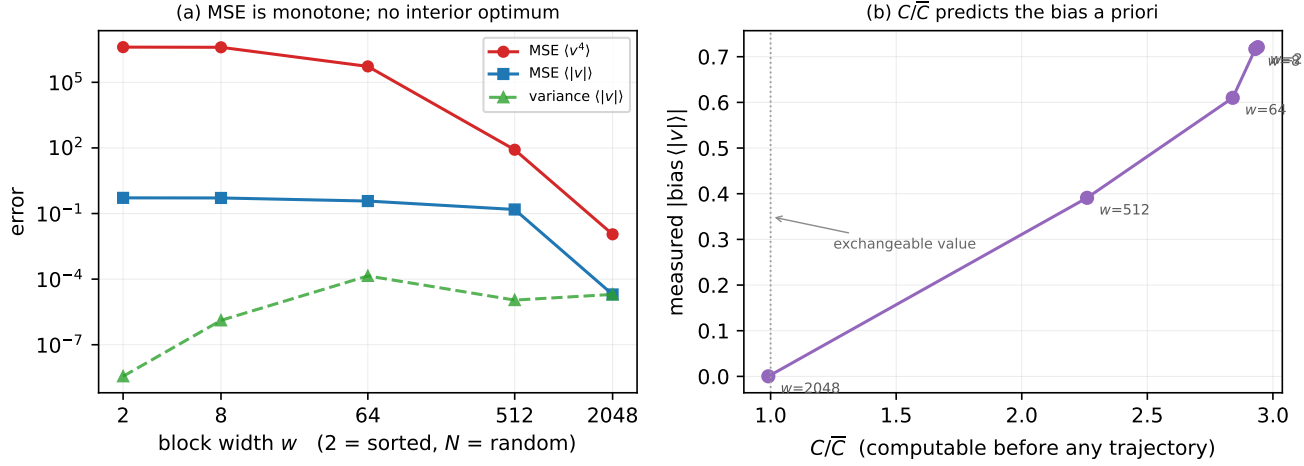


Figure 3: (a) Mean squared error against block width, aligned angles throughout, sixty randomizations. Monotone for both observables: no interior optimum. (b) The cross-term ratio  $C/\bar{C}$ , computable in  $O(N)$  from a single state before any trajectory is run, against the bias subsequently measured. Data generated by `kac_phasef_v2.1.0.py`; plotted by `kac_fig3_tradeoff_v1.0.0.py`.

Table 5: The implementation’s own sorting rule, evaluated on its own Maxwellian load.  $C/\bar{C}$  is deterministic given the load. The observed discrepancy is a window mean over the interval  $[20, 40]$ , as a ratio to the Monte Carlo floor.

| $N$  | $C/\bar{C}$ | predicted one-step $\Delta\mathbb{E}_\sigma[Q]/\mathbb{E}_\sigma[Q]$ | observed discrepancy ratio | absolute |
|------|-------------|----------------------------------------------------------------------|----------------------------|----------|
| 2048 | 2.713       | +43%                                                                 | 35.0                       | 0.632    |
| 8192 | 2.834       | +46%                                                                 | 72.3                       | 0.652    |

value by a factor near three, so the equilibrium is not preserved, and the fourth moment must rise by some forty percent on the first collision step.

The observed behaviour matches. The discrepancy sits at 35 and 72 times the Monte Carlo floor, and because the ratio grows as  $\sqrt{N}$  while the absolute discrepancy does not, the error is independent of the particle number: adding particles does not help. The velocity distribution reached is the condensate anticipated by Corollary 4.9, with mass accumulating near zero and a thin population carrying the energy.

Two features explain why this survived a good deal of investigation. The first is Table 2: the conserved moments are pinned, so the routine checks pass. The second is subtler. Across independent randomizations the discrepancy ratio at  $N = 2048$  came back as 35.21, 34.82 and 34.99, a spread of about half a percent, while the correct arm’s ratios spread by nearly thirty percent. The broken arm looks the more converged of the two. It is not: its error is roughly one hundred and seventy replicate standard deviations, so every replicate agrees closely on the same wrong answer, and no amount of replication will reveal the problem. **The tight agreement is the symptom.**

## 8 Which sorting strategies remain admissible

Theorem 4.8 indicts rules that bias  $C$  and is silent about the rest, and the silence is informative. The obstruction is not that a scheme sorts, but that it sorts on the velocities.

**Corollary 8.1.** *If the key by which particles are ordered or grouped carries no information about  $v$ , then the rule is state-independent as far as the collision dynamics is concerned and  $\sigma$  is preserved.*

*Proof.* Immediate from Proposition 2.3: the matching is then drawn from a distribution not depending on  $v$ .  $\square$

This covers the cell rules of Section 3, and with them the binary-collision schemes in standard use [14], which group by spatial position and match at random within a group. At equilibrium the velocities carry no information about position, so such schemes are admissible, and the theorem says why. It also explains why the difficulty has not been encountered before: a scheme that groups spatially gains nothing from velocity alignment and has never been asked to provide it.

The comparison with the setting where Array-RQMC has done best is instructive. In the walk on spheres solvers of Ho and Owen [5, 6] the walkers are independent. Each trajectory is a separate sample, and the sort decides only which low-discrepancy point a walker receives, leaving the law of every individual walk untouched. Sorting there can be as aggressive as one likes, and with a Hilbert sort it buys the large reductions those papers report. In a collision model the walkers are not independent. The sort decides which particles collide with which, so it enters the dynamics rather than the assignment of randomness, and the ordering that aligns the ensemble is the ordering that correlates collision partners. The distinction worth carrying away is therefore not between problems where Array-RQMC works and problems where it does not, but between sorts that assign randomness to independent trajectories and sorts that rewire an interaction.

In practice one wants to check a particular implementation rather than argue about it, and  $r(v)$  is the test, standing in for the  $r_\sigma$  that Theorem 4.8 constrains. Here a point of definition matters. The exchangeable value (6) averages over *all* matchings, whereas a rule that groups particles can only produce matchings within groups. That restriction is harmless in expectation: Lemma 4.6 gives  $\mathbb{E}_\sigma[C] = \mathbb{E}_\sigma[\overline{C}]$  for every state-independent rule, and by Corollary 8.1 a grouping key carrying no velocity information is state-independent, so such a rule should read one against the global reference with no dependence on the occupancy.

Measured on the loads of Section 7, a cell rule keyed on position instead reads  $0.94 \pm 0.01$  at occupancy 32 and  $0.98 \pm 0.01$  at occupancy 128, the quoted uncertainty being the spread across twelve independent seeds rather than the scatter within any one of them. The gap is not an artefact of the reference. Every cell in these loads holds an even number of particles, 32 and 128 respectively, so nothing goes unpaired and no occupancy effect arises. Two controls locate the cause. Replacing the position key by a random assignment of the same particles to cells of the same size returns  $1.00 \pm 0.01$ , and keeping the position key while permuting the velocities among the particles returns  $1.00 \pm 0.01$ ; both sit at unity at both particle numbers, as Lemma 4.6 requires. What the diagnostic is detecting is a correlation between position and velocity in the load itself.

The lesson is a practical qualification of Corollary 8.1. The corollary is a statement about the key, and position is velocity-free only in a load where the two are uncorrelated. In a plasma that has been evolved, or loaded with any spatial structure in its temperature, a cell index carries velocity information and the rule keyed on it is state-dependent after all. Whether a key is velocity-free is a property of the key together with the state it is applied to, and it should be measured rather than

assumed. Against the within-group reference

$$\overline{C}_{\text{cell}}(v) = \sum_c \frac{(\sum_{i \in c} v_i^2)^2 - \sum_{i \in c} v_i^4}{2(n_c - 1)}$$

the same configurations return  $1.00 \pm 0.01$  throughout. The two references answer different questions and both are worth computing. The within-group value conditions on the grouping, so it detects bias introduced *inside* the groups and is blind to whatever velocity information the grouping itself carries. The global value sees both together. A rule that reads one against the first and below one against the second, as this one does, is telling the practitioner that its pairing is innocent and its grouping is not.

These measurements are of one cell rule, at two particle numbers and two cell counts. They are consistent with Corollary 8.1 and were run to check it, not to establish a general claim about cell-based schemes: what licenses those is the corollary, and what the corollary requires is that the grouping key carry no velocity information. A scheme whose cells correlate with velocity, as they may away from equilibrium or under a driven load, is not covered.

The two cautions of Section 6 apply here unchanged: the test is one-sided, and  $r(v)$  is a sample quantity whose load-to-load spread sets the smallest difference worth reading.

## 9 Open problems

**Conjecture 9.1** (the stationary law). Theorem 4.8 fixes the direction in which the first collision moves  $\mathbb{E}_\sigma[Q]$  and says nothing about where the chain settles. We do not know the stationary law of a biased rule, even for the sorted rule at  $N = 4$ , where the state space is  $\mathcal{S} \subset \mathbb{R}^4$  and the dynamics may be tractable.

This is the weakest point of the present treatment. The numerical evidence of Section 6 describes the limit in some detail without identifying it.

**Conjecture 9.2** (length of the hierarchy). Is there a finite collection of moment conditions whose simultaneous satisfaction implies that  $\sigma$  is preserved? The construction of Proposition 5.3 defeats the fourth-moment condition alone using one free parameter, and a construction with  $k$  free parameters might be expected to defeat any  $k$  conditions. Proposition 5.2 shows the hierarchy is degenerate at  $N = 4$ , where the sixth-moment condition is implied by the fourth, so a second question is where it begins.

**Conjecture 9.3** (how much velocity information a sort may carry). Grouping the particles into  $N/w$  blocks by speed conveys  $\log_2(N/w)$  bits per particle about the velocities, while a full ordering conveys  $\log_2 N$ . At  $N = 2048$  we find one bit admissible and two not, against the eleven a full ordering supplies. Whether the admissible budget is a constant number of bits, a constant number of groups, or some fraction of  $\log_2 N$  cannot be determined from a single particle number.

Finally, the model treated here is the Kac caricature. Lemma 4.4 removes the restriction to uniformly resampled scattering angles, but the velocities remain one-dimensional and every pair collides at every step. Whether the obstruction survives in three dimensions, with variable collision rates and a physical cross section, is the question that decides how far the conclusion reaches. A second limit is narrower but nearer to practice. Corollary 8.1 clears any key carrying no velocity information, and that is a statement about the key together with the state it is applied to, not about

the key alone: in a load where position and velocity are correlated, a spatial key carries velocity information and the guarantee lapses. Section 8 measures exactly that happening. A practitioner relying on the corollary should therefore measure whether the chosen key is velocity-free on the state at hand rather than assume it from the key’s definition.

## 10 Conclusion

We began with a method that works and a problem that ought to suit it. Array-RQMC earns its variance reductions by sorting trajectories and handing them low-discrepancy points in that order, and collisional kinetics offers long chains, low per-step dimension, and an average over many particles. The obvious sorting key is the velocity, because the velocity is what the collision operator acts on.

That is exactly the key that cannot be used, once the order it induces is what selects the collision partners. Averaged over the scattering angle, the pairing enters the one-step fourth moment through one scalar (Lemma 4.2), and that scalar sits at a definite value for any rule chosen independently of the state. A rule that correlates the velocities of collision partners moves it, and the equilibrium moves with it, at every even  $N \geq 4$  and under any symmetric scattering law shared by all pairs. The alignment that buys the variance reduction and the correlation that destroys the equilibrium are not two effects to be balanced against each other. They are one structure seen twice, which is why Figure 3(a) shows no interior optimum: there is nothing to tune.

Three things follow that we would want a practitioner to take away. Sorting is not the difficulty; sorting on velocity is, and a grouping key carrying no velocity information leaves the equilibrium intact. The cross term can be computed in one pass before any trajectory is run, so a proposed scheme can be tested in advance rather than diagnosed afterwards. And the failure does not announce itself: the conserved moments stay exact, so the usual checks pass, while the replicates agree with one another to a fraction of a percent around an answer that is wrong by many standard deviations. In this setting the close agreement of independent runs is not evidence that a computation has converged. It is the signature of the thing that has gone wrong.

What we have not settled is where the chain goes instead. The obstruction fixes the direction in which the first collision moves the fourth moment and says nothing about the stationary law it reaches (Conjecture 9.1). That distribution, seen clearly enough in Section 6 to be described and not clearly enough to be identified, is the natural next question.

## Acknowledgements

This work used generative AI assistance (Anthropic Claude). Its contribution spanned the formulation and checking of the proofs, the software used for the numerical experiments, the generation and processing of the data reported in Tables 3–5 and Figures 1–3, drafting of the exposition, and preparation of the bibliography. Every proof was verified independently of the tool; every figure and table is produced by code from stored data rather than by a generative process, and that code is available with the paper; every cited work has been read by the author and its bibliographic details checked against primary sources. **The author assumes responsibility for all content.**

# References

- [1] Russel E. Caflisch, William Morokoff, and Art B. Owen. Valuation of mortgage backed securities using Brownian bridges to reduce effective dimension. *Journal of Computational Finance*, 1(1):27–46, 1997.
- [2] Josef Dick and Friedrich Pillichshammer. *Digital Nets and Sequences: Discrepancy Theory and Quasi-Monte Carlo Integration*. Cambridge University Press, Cambridge, 2010.
- [3] Mathieu Gerber and Nicolas Chopin. Sequential quasi Monte Carlo. *Journal of the Royal Statistical Society, Series B*, 77(3):509–579, 2015.
- [4] Zhijian He and Art B. Owen. Extensible grids: uniform sampling on a space filling curve. *Journal of the Royal Statistical Society, Series B*, 78(4):917–931, 2016.
- [5] Valerie N. P. Ho and Art B. Owen. Randomized quasi-Monte Carlo for walk on spheres. Technical Report arXiv:2605.08483, arXiv, 2026.
- [6] Valerie N. P. Ho and Art B. Owen. Walk on spheres and Array-RQMC. Technical Report arXiv:2605.12844, arXiv, 2026.
- [7] Mark Kac. Foundations of kinetic theory. In Jerzy Neyman, editor, *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, Volume III*, pages 171–197. University of California Press, Berkeley and Los Angeles, 1956.
- [8] Christian Lécot and Bruno Tuffin. Quasi-Monte Carlo methods for estimating transient measures of discrete time Markov chains. In Harald Niederreiter, editor, *Monte Carlo and Quasi-Monte Carlo Methods 2002*, pages 329–343. Springer, Berlin, 2004.
- [9] Pierre L’Ecuyer, Christian Lécot, and Bruno Tuffin. A randomized quasi-Monte Carlo simulation method for Markov chains. *Operations Research*, 56(4):958–975, 2008.
- [10] Pierre L’Ecuyer, David Munger, Christian Lécot, and Bruno Tuffin. Sorting methods and convergence rates for array-RQMC: some empirical comparisons. *Mathematics and Computers in Simulation*, 143:191–201, 2018.
- [11] Ruixue Liu and Art B. Owen. Estimating mean dimensionality of analysis of variance decompositions. *Journal of the American Statistical Association*, 101(474):712–721, 2006.
- [12] Harald Niederreiter. *Random Number Generation and Quasi-Monte Carlo Methods*. Number 63 in CBMS-NSF Regional Conference Series in Applied Mathematics. SIAM, Philadelphia, 1992.
- [13] Art B. Owen. Practical quasi-monte carlo integration, 2023. <https://artowen.su.domains/mc/practicalqmc.pdf>.
- [14] Tomonori Takizuka and Hirotada Abe. A binary collision model for plasma simulation with a particle code. *Journal of Computational Physics*, 25(3):205–219, 1977.